

# Swasti S. Mishra

Amsterdam, Netherlands | Email | Homepage | Scholar | GitHub | LinkedIn

## INTERESTS

Hyperbolic and non-Euclidean deep learning for vision, leveraging the hierarchy, robustness, and interpretability these geometries naturally provide. Also, controlled video and image generation.

## EDUCATION

|                                                                                                                                                                              |                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>University of Amsterdam (UvA)</b><br><i>Ph.D. Candidate in Computer Vision (HAVA Lab)</i><br>Advisors: Dr. Pascal Mettes and Prof. Dr. Erwin Berkhout                     | June 2024 - Expected June 2028<br>Amsterdam, Netherlands |
| <b>International Institute of Information Technology Bangalore (IIITB)</b><br><i>Integrated Master of Technology in Computer Science Engineering; Cumulative GPA: 3.75/4</i> | August 2017 - July 2022<br>Bangalore, India              |

## PUBLICATIONS & PATENTS

- [1] Daniel Uyterlinde, **Swasti S. Mishra**, Pascal Mettes. **Hyperbolic Concept Bottleneck Models**. In *Neural Information Processing Systems (NeurIPS 2026, Spotlight)*. [link]  
→ *Mentored & Supervised 1 Master's student from project formulation to Spotlight acceptance.*
- [2] **Swasti S. Mishra**, Max van Spengler, Erwin Berkhout, Pascal Mettes. **Hyperbolic U-Net for Robust Medical Image Segmentation**. In *Medical Imaging with Deep Learning (MIDL 2026)*. [link]  
→ *Awarded MIDL Best Poster Pitch Award.*
- [3] Max Wilde, Mattia Sferrazza, Selman Gul, **Swasti S. Mishra**. **No Free Robustness: On Bolting Hyperbolic Heads onto Frozen Vision Foundation Models**. In *European Conference on Computer Vision Workshops (ECCVW 2026)*. [link]  
→ *Supervised a team of graduate students for a Foundation Models course, leading to empirical benchmark findings.*
- [4] **Swasti S. Mishra**, Kuldeep Kulkarni, Duygu Ceylan Aksit, Balaji Vasani Srinivasan. **Attention map correction for garment animation generation**. *US Patent (2026)*.
- [5] **Swasti S. Mishra**, Tripti Shukla, Srikrishna Karanam, Balaji Vasani Srinivasan. **User Content Conditioning in Graphic Design Layout Generation**. *US Patent (2026)*.
- [6] Kasarla et al. (incl. **Swasti S. Mishra**). **HierVision: Standardized and Reproducible Hierarchical Sources for Vision Datasets**. In *International Conference on Computer Vision Workshops (ICCVW 2025)*. [link]
- [7] Goyal et al. (incl. **Swasti S. Mishra**). **Design-o-meter: Towards evaluating and refining graphic designs**. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025)*. [link]
- [8] Renier et al. (incl. **Swasti S. Mishra**). **A deepfake-based study on facial expressiveness and social outcomes**. In *Scientific Reports*, 14(1), 3642. (2024). [link]
- [9] Agarwal et al. (incl. **Swasti S. Mishra**). **SketchBuddy: Context-Aware Sketch Enrichment and Enhancement**. In *ACM Multimedia Systems Conference (ACM MMSys 2023)*. [link]
- [10] **Swasti S. Mishra**, Kumar Shubham, and Dinesh Babu Jayagopi. **A Hybrid Rigid and Non-Rigid Motion Approximation for Generating Realistic Listening Behavior Videos**. In *Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP 2022)*. [link]

## EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Ph.D. Candidate, University of Amsterdam</b><br><ul style="list-style-type: none"><li>• <b>GPU Acceleration &amp; Kernels:</b> Accelerated hyperbolic tensor operations and training/inference throughput while preserving numerical precision by engineering custom Triton kernels.</li><li>• <b>Architecture &amp; Robustness:</b> Achieved state-of-the-art segmentation robustness over Euclidean baselines across 7 medical datasets under 6 noise distribution shifts by designing a fully Hyperbolic U-Net architecture with Newton's approximation weight initialization.</li><li>• <b>Applied Dental Imaging:</b> Benchmarked training strategies for fusing multiple dental modalities for caries diagnosis. Generated synthetic dental images and ran a pilot reader study in which dentists judged synthetic bitewing radiographs as real or AI-generated, evaluating their use for model training and dental education (both ongoing).</li></ul> | June 2024 - Present |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|

- **Research Leadership & Supervision:** Supervised and mentored 5+ Master's and graduate students across geometric deep learning and foundation model adaptation projects, leading to 2 top-venue workshop/conference papers.

**Research Associate, Adobe Inc.**

July 2022 - April 2024

- **Video Diffusion Models:** Enhanced video temporal coherence across text-to-video models (ControlNet, Stable Diffusion, Adobe Firefly) by formulating a flow-warping self-attention mechanism using optical flow from normal maps, yielding **1 granted US Patent**.
- **Layout AI & Graphic Design:** Developed context-aware design evaluation and conditional layout generation models, resulting in **1 granted US Patent**.

**Research Intern, Awi Japan**

January 2022 - April 2022

- Generated synthetic human-object tracking datasets using GTA V modding and adapted single-shot joint detection and embedding networks for real-time tracking.

**Mitacs Globalink Research Intern, University of Waterloo**

August 2021 - October 2021

- Improved flood risk estimation precision on Google Street View API datasets by integrating depth feature-assisted multi-modal deep learning architectures.

**Research Intern, Adobe Inc.**

May 2021 - August 2021

- Developed context-aware sketch enrichment tools enabling rapid lo-fi UI/UX prototyping (published at ACM MMSys 2023).

**Research Intern, TU Dresden**

May 2020 - July 2020

- Integrated LDA2Vec text mining models and Neo4j graph databases to search lipid signatures in biological pathways.

**Software Engineering Intern, Microsoft Corp.**

June 2019 - July 2019

- Developed automated interview scheduling applications using React Native, Django, and Azure Graph APIs.

## ACHIEVEMENTS & ACADEMIC SERVICE

---

- **Silver Reviewer Award:** Recognized for outstanding review quality at ICML (2026).
- **Peer Reviewer:** Invited peer reviewer for ICML, NeurIPS, AAAI, TMLR, IJCV.
- **Adobe Women in Technology Scholarship:** Awarded to top 6 female computer science scholars in India (2021).
- **Academic Honors:** Late Sri. N. Rama Rao Medal for All-Rounder of the Year (IIITB, 2022); Dean's Merit List (2017–2022).
- **Competitive Programming & Hackathons:** Qualified for ACM-ICPC India Regionals (2018); 1st place in Android Track at Google 'Build For India' Hackathon (2018).

## TECHNICAL SKILLS

---

- **Languages & Frameworks:** Python, PyTorch, Triton, C++ (familiar).
- **Tools & Infrastructure:** SLURM, Weights & Biases, Git.
- **Domains:** Computer Vision, Non-Euclidean/Hyperbolic Deep Learning, Diffusion Models, Generative AI, Vision Foundation Models, Medical Image Analysis, Custom GPU Kernel Optimization (Triton).